Pre Charge (off board)

+48V_BATT
J3 Conn_Batt_PWR
J15 Conn_Batt_GND
PGND
48V_FUSE
F1 Fuse
F9 MF-MSMF010-2
RPRE1 100
D1 TVS
D2 TVS
SMCJ58A TVS Diode Hard Overvoltage Protection
WRONG VOLTAGE FOR TVS
PRECHM
J13
PRECHOUT
J14
Precharge Ctrl In
J24
24V_DIS
PRECH_EN
+3.3V
J22 Precharge Contactor Coil Control
C_FUSE
Q1 NMOS
DFB1 S1M
RG1 100
RPDQ1 100k
12V_BIAS
U15 MCP1402T-E/OT
GND VDD OUT GND IN
C0RV1 1uF
RPDP1 100k

Main rail bulk caps

CBLK3 470u
CBLK4 470u
CBLK5 470u
CBLK6 470u
CBLK7 470u
CBLK8 470u
48V_PRECH

Main Current Sensor

U4 ACS758LCB-100U-PFF-T
+3.3V
CCS2 100n
RCSF2 1k
Main_curr
CCSF2 10n
48V_MAIN
J/ESTOP1 ESTOP_IN
AGND

Back current protection

48V_ESTOP
RFBC1 1.65M
RDIV1 1M
RDIV2 24k
U1 TLV3011AIDBVT
Trip point: 54V
Release: 51V
Peak: 54.5
Change RFBC1 For thighter Trip and Release
RPU1 10k
U6 UCC27511DBVR
CCS3 100n
+6V
RFB1 4.7R
RFB2 4.7R
J16 Dump_Resistor_In
J17 Dump_Resistor_Out
DFB2 D
Q6 NMOS
J18 BC_Disconnect
RPD1 100k

ESTOP

Notes for ESTOP:
- ESTOP Button NEEDS to be a normally closed type.
- ESTOP Button in series w/ contactor coil
- Both fets are in series for double fault protection.
- Fet Gates are controlled by 3.3V --> 12V by driver IC

J20 ESTOP Contactor Coil Control
J21 ESTOP Button (Normally Closed)
FB 1A Fuse
Varistor ZV50K2220801NIR1
Q2 NMOS IRFR220N
Q3 NMOS IRFR220N
RESTP5 100k
RESTP8 100k
RESTP9 100k
RPD2 100k
RESTP7 4.7K 1W
RESTP6 7.5k
RESTP1 10
RESTP2 10
D9 12V Zener 1W
D11 3.3V Zener
CCS10 100nF 50V
CCS11 100nF 50V
CCS12 100nF 50V
Should be placed as close as possible to gate drivers
Pulldowns in case 12 V rail goes down
These resistors are for dampening the ringing from driver ICs a bit
U13 MCP1402T-E/OT
U14 MCP1402T-E/OT
12V "RAIL"
ESTOP_EN1
ESTOP_EN2
RESTP4 2.2k
RESTP3 2.2k

Actuator rails

48V_ESTOP
U8 ACS758LCB-100U-PFF-T
RCSF4 1k
Act_rail_curr
CCS4 100n
CCSF4 10n
JESTOP2 ESTOP_OUT
48V_ACT
F7
48V_ACT1_FUSE
CBLKA1 100uF
CINC1 4.7u
CINC2 4.7u
CINC9 100nF
48V_ACT1
D5 TVS
J8 Conn_01x02
U10 ACS758LCB-100U-PFF-T
CCS5 100n
RCSF5 1k
CCSF5 10n
F6
48V_ACT2_FUSE
CBLKA2 100uF
CINC3 4.7u
CINC4 4.7u
CINC10 100nF
48V_ACT2
D4 TVS
J7 Conn_01x02
U9 ACS758LCB-100U-PFF-T
CCS6 100n
RCSF6 1k
CCSF6 10n
F5
48V_ACT3_FUSE
CBLKA3 47uF
CINC5 4.7u
CINC6 4.7u
CINC11 100nF
48V_ACT3
D7 TVS
J10 Conn_01x02
U12 ACS758LCB-100U-PFF-T
CCS7 100n
RCSF7 1k
CCSF7 10n
F4
48V_ACT4_FUSE
CBLKA4 47uF
CINC7 4.7u
CINC8 4.7u
CINC12 100nF
48V_ACT4
D6 TVS
J9 Conn_01x02
LOOK AT BIDIRECTIONAL VERSION
U11 ACS758LCB-100U-PFF-T
CCS8 100n
RCSF8 1k
CCSF8 10n
Main_curr
Com_cur
J11 Conn_01x08

48V to 3.3V buck

48V_MAIN
U3 LMR36015ARNXR
400KHZ SF
CINA1 47uF
CINA2 4.7uF
CINA3 220nF
CHF4A1
CINSA1 100nF
CVCCA1 1uF
AGNDA5
CBOOT2 100nF
L1 18uH 28.45% ripple
COUT1A1 47uF
COUT2A1 47uF
COUT3A1 100nF
CFF1 20pF
RENT1 100k
RENB1 43.2k
+3.3V
J4 3.3V Out

48V to 24V buck + current sensor

48V_MAIN
F2
CIN1B1 47uF
CIN2B1 4.7u
CIN3B1 4.7u
CIN4B1 4.7u
CIN5B1 100nF
24V_FUSE
RUV1 499k
Needs to be connected to precharge signal connector
24V_EN
RUV2 12.7k
CEN1 1nF
24V_DIS
RGEN1 100k
Q9 2N7002
AGND2
RCON1 41.2k
RSF1 47k
U2 LM5148RGYR
SF 461.25 kHz
CVCC1 2.2u
CVDDA1 100n
CBOOT1 100n
Q4 NVMFS6H858NLT1G
Q5 NVMFS6H848NLT1G
24V_NMOS
32.6% ripple
L2 10uH
RSEN1 0.002R
MAY NEED TO GO TO 4-5 mR
24V_IND
RCOMP1 8.25k
CCOMP1 4.7nF
CHF1
Validate and choose CHF1.
Calculate phase margin
RFB1A2 291k
RFB2B2 10k
COUT1B1 22uF
COUT2B1 22uF
COUT3B1 22uF
COUT4B1 100nF
24V_SEN
U5 ACS758LCB-100U-PFF-T
CCS1 100n
RCSF1 1k
CCSF1 10n
Com_cur
D3 TVS
J5 24V Out --> Jetson
J6 24V Out --> USB-C-PD controller
+24V

48V to 6V Converter

48V_ACT
F3
6V_FUSE
CIN1B2 100uF
CIN2B2 4.7u
CIN3B2 4.7u
CIN4B2 4.7u
CIN5B2 100nF
U7 LM5148RGYR
CVCC2 2.2u
CVDDA2 100n
CBOOT3 100n
RCON2 41.2k
RSF2 47k
SF 461.25 kHz
Q7 NVMFS6H858NLT1G
Q8 NVMFS6H848NLT1G
6V_NMOS
+6V_IND
L3 4.7uH
-30.4% ripple
RSEN2 0.002R
MAY NEED TO GO TO 4-5 mR
D8 TVS
+6V
COUT1B2 22uF
COUT2B2 22uF
COUT3B2 22uF
COUT4B2 100nF
CBULK2 100uF
J12 6V Out
RFB1A1 64.9k
RFB2B1 10k
RCOMP2 6k
CCOMP2 5.6nF
CHF2 100pF
Validate and choose CHF1 (suspected to be 22pF.
Calculate phase margin

Thermistors

Thermistors
Final will include more than 3.
+3.3V
J1 Conn_01x03
R1 10k
R3 10k
R6 10k
TH1 Thermistor_NTC
TH2 Thermistor_NTC
TH3 Thermistor_NTC
Thermistors to be placed nearby FETs.
Negative Temperature Coeffecient resistors so disconnect reads high.
AGND

Power flags and net ties

+48V_BATT PWR_FLAG
+3.3V PWR_FLAG
+6V PWR_FLAG
+12V PWR_FLAG
PGND NT1 NetTie_2 AGND1
PGND NT2 NetTie_2 AGND2
PGND NT3 NetTie_2 AGND3
PGND NT4 NetTie_2 AGND4
PGND NT5 NetTie_2 AGND5
PGND NT6 NetTie_2 AGND6
PGND NT7 NetTie_2 AGND7
PGND NT8 NetTie_2 AGND8
PGND NT9 NetTie_2 AGND9
PGND NT11 NetTie_2 AGND10
PGND NT10 NetTie_2 AGND

Mounting holes

H1 MountingHole_3.2mm_M3
H2 MountingHole_3.2mm_M3
H3 MountingHole_3.2mm_M3
H4 MountingHole_3.2mm_M3

J2 Conn_01x04
ESTOP_EN1
ESTOP_EN2
ESTOP_FB
PGND

WiscoHumanoids
Sheet: /
File: Main_PDB.kicad_sch
Title: PDB
Size: A1
Date: 2026-09-09
KiCad E.D.A. 10.0.0
Rev:
Id: 1/1